

Do the Quantum Parts Help?

A Component-Wise Ablation of the Hamiltonian Vision Kernel

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Overview

1. Motivation
2. HVK Pipeline (recap)
3. Evaluation Protocol
4. Results: Component Ablation
5. Results: Held-Out Real Images
6. The Leakage Lesson
7. Scoped Positive Results
8. Conclusion

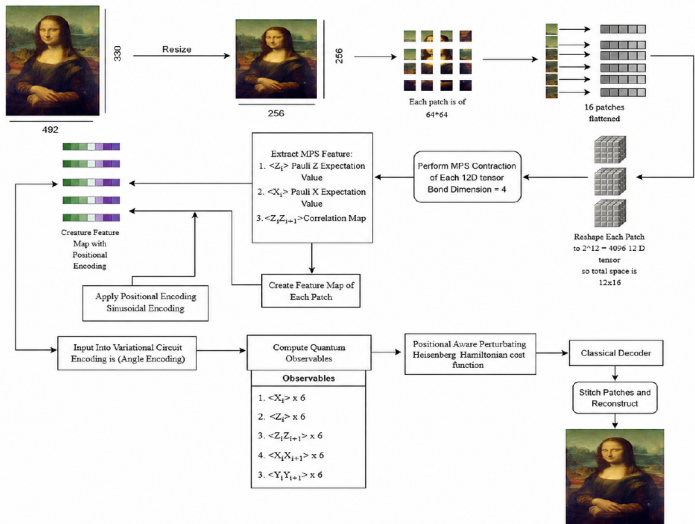
Motivation

- Hybrid quantum–classical vision models routinely claim an advantage over classical baselines.
- Rarely isolated: **which component** is actually responsible?
- Rarely matched: classical control parameter budget vs. quantum model.
- Rarely audited: same-set fit vs. held-out generalization; target–feature leakage.

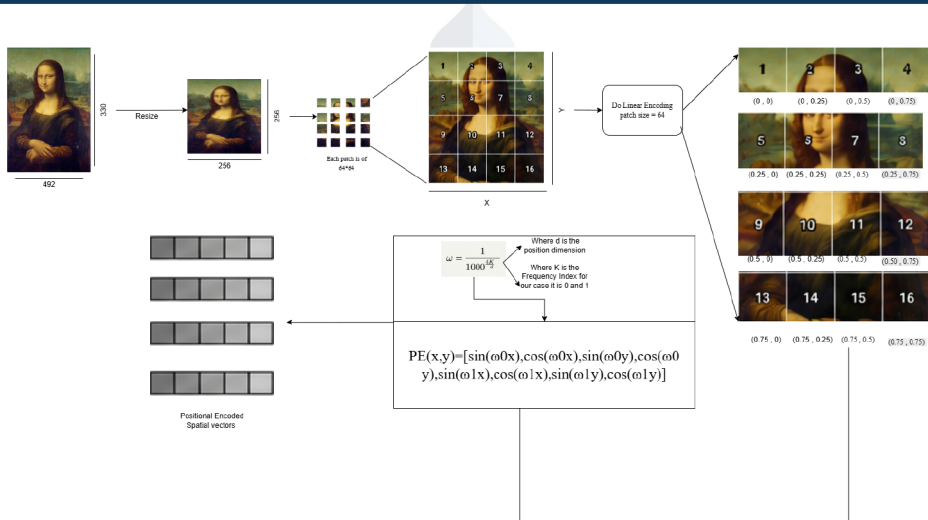
Question

Does *any* quantum component of HVK do measurable work on real, held-out images?

Model Pipeline



Positional Encoding



Evaluation Protocol

- **Freeze isolation:** train only classical decoder, or only quantum circuit.
- **Resource-matched controls:** same feature width, same readout parameter count.
- **Leakage audit:** target must not be linearly recoverable from a model's own input features.
- **Multi-seed statistics:** 5 seeds, paired Wilcoxon test, bootstrap 95% CI.
- **Held-out $\neq$ same-set:** generalization claims require held-out splits only.

Which Component Is Load-Bearing? (Monalisa, 5 seeds)

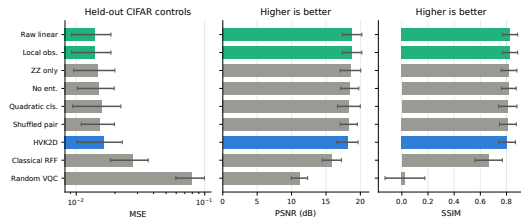
Variant	PSNR (dB)	Δ PSNR	Verdict
Freeze classical, train quantum	10.93 ± 0.00	-22.03	Collapses
Random VQC output	27.96 ± 0.28	-5.01	Much worse
Shared VQC baseline	32.97 ± 0.22	0.00	Reference
No entanglement	32.98 ± 0.15	$+0.02$	Not significant
No MPS	33.04 ± 0.15	$+0.08$	Not significant
Freeze quantum, train classical	33.26 ± 0.10	$+0.30$	Ties baseline
Parameter-matched classical	33.37 ± 0.08	$+0.40$	Exceeds baseline
No energy loss	33.40 ± 0.06	$+0.44$	Exceeds baseline
Classical tanh replacement	33.50 ± 0.00	$+0.54$	Exceeds baseline

The classical decoder is load-bearing. The quantum circuit is not.

Headline Result: Held-Out CIFAR-10

Model	PSNR (dB)	Paired
Local observables / raw-linear	18.80 ± 1.42	
ZZ-only observables	18.56 ± 1.49	
No entanglement	18.46 ± 1.32	
HVK2D (trained)	18.12 ± 1.54	
Random VQC	11.15 ± 1.19	

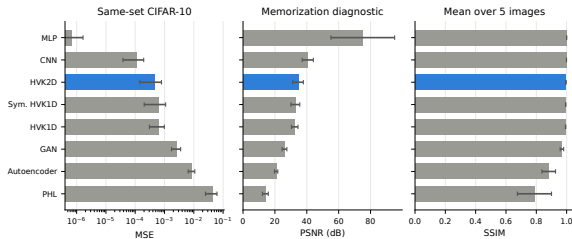
Wilcoxon vs. raw-linear: $p = 9.5 \times 10^{-6}$, $n = 20$



Every quantum ablation ties or beats trained HVK2D on held-out images.

Same Pattern Across Six Datasets

Dataset	Best local/raw	HVK2D pair	Winner
CIFAR-10 (native)	21.17 ± 1.98	19.99 ± 2.02	Local/raw
MNIST	16.55 ± 1.27	16.12 ± 1.34	Local/raw
Fashion-MNIST	17.67 ± 1.71	17.33 ± 1.80	Local/raw
PathMNIST	27.52 ± 6.02	27.01 ± 5.81	Local/raw
BloodMNIST	24.79 ± 1.78	24.24 ± 1.66	Local/raw
PneumoniaMNIST	26.49 ± 2.48	25.73 ± 2.29	Local/raw



A Circular Benchmark — and How We Caught It

- An earlier diagnostic reported $R^2 = 1.0000$, PSNR = 120 dB for the entangling model.
- Cause: the target was built from the **same product terms** handed to the model as input features.
- The model was copying, not learning. Withdrawn from every claim in the paper.

Reusable rule

Before trusting a favorable diagnostic: verify the target is *not* linearly recoverable from a candidate's own input features.

Two Narrow, Honest Positive Results

Entanglement-necessity

Model	R^2
HVK2D entangling	0.974
No entanglement	0.019
Random VQC	-0.173

Deliberately favorable,

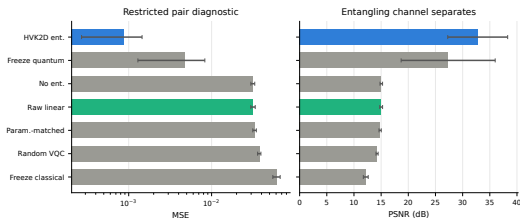
leakage-audited synthetic target.

D_4 symmetry (provable)

Map	Equiv. error
D_4 -pooled HVK2D	9.6×10^{-17}
HVK2D positional	0.815
Local/raw	0.837

Verified over 7000

patch transforms.



Guidelines for Evaluating Hybrid Quantum–Classical Vision Models

1. Resource-match every classical control.
2. Freeze-isolate each trainable component.
3. Audit every favorable diagnostic for target–feature leakage.
4. Report held-out, not same-set, performance as primary evidence.
5. Report multiple seeds with error bars.

Conclusion

- On held-out real images, **no quantum component of HVK is load-bearing** — classical controls tie or win.
- The classical decoder, not the quantum circuit, does the work.
- Entanglement helps only in a narrow, leakage-audited, pair-correlation setting.
- D_4 -pooling gives a provable symmetry guarantee, independent of any advantage claim.

The contribution is the evaluation protocol itself.

THANK YOU!

